perfectly predictably: 5.1, 5.2, 5.3, and so on. A wave function's numbers can go up and down again, like a wave—hence the name—but they'll always undulate smoothly like waves too, never jerking around too crazily.

Wave functions aren't too complicated, but it's a little weird that quantum physics needs them. Newton could give you the location of any object using just three numbers. Apparently, quantum physics needs an infinity of numbers, scattered across the universe, just to describe the location of a single electron. But maybe electrons are weird—maybe they don't behave the way that rocks or chairs or people do. Maybe they're smeared out, and the wave function describes how much of the electron is in a particular place.

But, as it turns out, that can't be right. Nobody's ever seen half of an electron, or anything less than a whole electron in one well-defined place. The wave function doesn't tell you how much of the electron is in one place—it tells you the *probability* that the electron is in that place. The predictions of quantum physics are generally in terms of probabilities, not certainties. And that's strange, because the Schrödinger equation is totally deterministic—probability doesn't enter into it at all. You can use the Schrödinger equation to predict with perfect accuracy how any wave function will behave, forever.

Except that's not quite true either. Once you do find that electron, a funny thing happens to its wave function. Rather than following the Schrödinger equation like a good wave function, it collapses—it instantly becomes zero everywhere except in the place where you found the electron. Somehow, the laws of physics seem to behave differently when you make a measurement: the Schrödinger equation holds all the time, except when you make a measurement, at which point the Schrödinger equation is temporarily suspended and the wave function collapses everywhere except a random point. This is so weird that it gets a special name: the *measurement problem* (Figure 1.1).